

Linear Algebra Problem Set #1

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Q1

1(a)

$$\begin{aligned}(AB)^T &= \left(\begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix} \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix} \right)^T = \left(\begin{bmatrix} 14+0 & 4+0 \\ 21+48 & 6+24 \end{bmatrix} \right)^T \\ &= \left(\begin{bmatrix} 14 & 4 \\ 69 & 30 \end{bmatrix} \right)^T = \begin{bmatrix} 14 & 69 \\ 4 & 30 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}B^T A^T &= \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}^T \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}^T = \begin{bmatrix} 7 & 6 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} 14+0 & 21+48 \\ 4+0 & 6+24 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 69 \\ 4 & 30 \end{bmatrix}\end{aligned}$$

1(b)

$$(AB)^{-1} = \begin{bmatrix} 14 & 4 \\ 69 & 30 \end{bmatrix}^{-1} = \frac{1}{144} \begin{bmatrix} 30 & -4 \\ -69 & 14 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{9} \begin{bmatrix} 3 & -2 \\ -6 & 7 \end{bmatrix} \cdot \frac{1}{16} \begin{bmatrix} 8 & 0 \\ -3 & 2 \end{bmatrix} = \frac{1}{144} \begin{bmatrix} 30 & -4 \\ -69 & 14 \end{bmatrix}$$

1(c)

$$(A^T)^{-1} = \left(\begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}^T \right)^{-1} = \left(\begin{bmatrix} 2 & 3 \\ 0 & 8 \end{bmatrix} \right)^{-1} = \frac{1}{16} \begin{bmatrix} 8 & -3 \\ 0 & 2 \end{bmatrix}$$

$$(A^{-1})^T = \left(\begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}^{-1} \right)^T = \left(\frac{1}{16} \begin{bmatrix} 8 & 0 \\ -3 & 2 \end{bmatrix} \right)^T = \frac{1}{16} \begin{bmatrix} 8 & -3 \\ 0 & 2 \end{bmatrix}$$

Q2

This is not the case, in general. Consider the counterexample

$$\det(\mathcal{I}_n) + \det(\mathcal{I}_n) = 2 \neq 2^n = \det(\mathcal{I}_n + \mathcal{I}_n)$$

Q3

3(a)

The Hessian is

$$\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

which has principal minors -2 and 3 . Therefore it's negative definite and therefore it's strictly concave.

3(b)

The Hessian is

$$\begin{bmatrix} 6 & -2 & 0 \\ -2 & 4 & 2 \\ 0 & 2 & 2 \end{bmatrix}$$

It's easy enough to see that its first two principal minors are 6 and 20 . The determinant is 16 . Therefore, the matrix is positive definite and the function is strictly convex.

Q4

Symmetry

To show that $M_{\mathbf{X}}$ is symmetric,

$$\begin{aligned}
\mathcal{M}_{\mathbf{X}}^T &= (\mathcal{I}_n - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T)^T \\
&= \mathcal{I}_n^T - (\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T)^T && \text{by the distributive property} \\
&= \mathcal{I}_n - (\mathbf{X}^T)^T (\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1})^T && \text{by the reverse-order law and } \mathcal{I}_n^T = \mathcal{I}_n \\
&= \mathcal{I}_n - \mathbf{X}((\mathbf{X}^T \mathbf{X})^{-1})^T \mathbf{X}^T && \text{by the reverse-order law and } (A^T)^T = A \\
&= \mathcal{I}_n - \mathbf{X}((\mathbf{X}^T \mathbf{X})^T)^{-1} \mathbf{X}^T && \text{since the inverse and transpose commute} \\
&= \mathcal{I}_n - \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T && \text{by } (\mathbf{X}^T \mathbf{X})^T = \mathbf{X}^T \mathbf{X} \\
&= \mathcal{M}_{\mathbf{X}}
\end{aligned}$$

Because $\mathcal{M}_{\mathbf{X}}^T = \mathcal{M}_{\mathbf{X}}$, the matrix $\mathcal{M}_{\mathbf{X}}$ is symmetric.

Idempotency

To show that $\mathcal{M}_{\mathbf{X}}$ is idempotent, write it as a difference of two matrices,

$$\mathcal{M}_{\mathbf{X}} = \mathbf{A} - \mathbf{B}, \quad \text{where } \mathbf{A} = \mathcal{I}_n \text{ and } \mathbf{B} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T.$$

Each of the four products appearing in $(\mathbf{A} - \mathbf{B})^2$ simplifies:

$$\begin{aligned}
\mathbf{A}^2 &= \mathcal{I}_n \mathcal{I}_n = \mathcal{I}_n = \mathbf{A} \\
\mathbf{AB} &= \mathcal{I}_n \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{B} \\
\mathbf{BA} &= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathcal{I}_n = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{B} \\
\mathbf{B}^2 &= \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \underbrace{\mathbf{X}^T \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1}}_{=\mathcal{I}} \mathbf{X}^T = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{B}
\end{aligned}$$

Expanding the square and substituting these four results gives

$$\begin{aligned}
\mathcal{M}_{\mathbf{X}}^2 &= (\mathbf{A} - \mathbf{B})^2 \\
&= \mathbf{A}^2 - \mathbf{AB} - \mathbf{BA} + \mathbf{B}^2 \\
&= \mathbf{A} - \mathbf{B} - \mathbf{B} + \mathbf{B} \\
&= \mathbf{A} - \mathbf{B} \\
&= \mathcal{M}_{\mathbf{X}}
\end{aligned}$$

so $\mathcal{M}_{\mathbf{X}}$ is idempotent.

Q5

The characteristic polynomial is

$$\begin{aligned}
(.8 - \lambda)(.95 - \lambda) - (.05)(.2) &= .76 - 1.75\lambda + \lambda^2 - .01 \\
&= \lambda^2 - 1.75\lambda + .75 \\
&= \frac{1}{4}(4\lambda^2 - 7\lambda + 3) \\
&= \frac{1}{4}(\lambda - 1)(4\lambda - 3)
\end{aligned}$$

Therefore, the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = \frac{3}{4}$

For $\lambda_1 = 1$:

$$\begin{aligned}
(C - \lambda_1 I)u &= 0 \\
\begin{bmatrix} 0.8 - 1 & 0.05 \\ 0.2 & 0.95 - 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} &= 0 \\
-0.2u_1 + 0.05u_2 &= 0 \\
u_2 &= 4u_1
\end{aligned}$$

Normalizing the vector, we get the eigenvector for $\lambda_1 = 1$ is: $u = \begin{bmatrix} \frac{1}{\sqrt{17}} & \frac{4}{\sqrt{17}} \end{bmatrix}$

For $\lambda_2 = \frac{3}{4} = 0.75$:

$$\begin{aligned}
(C - \lambda_2 I)u &= 0 \\
\begin{bmatrix} 0.8 - 0.75 & 0.05 \\ 0.2 & 0.95 - 0.75 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} &= 0 \\
0.05u_1 + 0.05u_2 &= 0 \\
u_2 &= -u_1
\end{aligned}$$

Normalizing the vector, we get the eigenvector for $\lambda_2 = 3/4$ is: $u = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$

Q6

The sum given in (a) is $\mathbf{u}'(\frac{1}{2}\Sigma^{-1})\mathbf{u}$ and the one given in (b) is $\mathbf{u}'\Lambda\mathbf{u}$.

Q7

The identities given in (a) follow directly from the definition of matrix multiplication.

The identity in (b) results from applying the identities in (a) to the identity $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = (\frac{1}{n} \mathbf{X}^T \mathbf{X})^{-1} \frac{1}{n} \mathbf{X}^T \mathbf{y}$ which follows from the commutativity and associativity of scalar-matrix multiplication and the identity $(\frac{1}{n} \mathbf{X})^{-1} = n \mathbf{X}^{-1}$

which can be obtained by applying the commutativity and associativity of scalar-matrix multiplication to show that $(\frac{1}{n}\mathbf{X})^{-1}n\mathbf{X}^{-1} = \mathcal{I}$.

(c) We have:

$$\begin{aligned}\hat{\beta} &= (X^T X)^{-1} X^T Y \\ \hat{\beta} &= (X^T X)^{-1} X^T (X\beta + u) \\ \hat{\beta} &= (X^T X)^{-1} X^T X\beta + (X^T X)^{-1} X^T u \\ \hat{\beta} &= I\beta + (X^T X)^{-1} X^T u \\ \hat{\beta} - \beta &= (X^T X)^{-1} X^T u \\ \sqrt{n}(\hat{\beta} - \beta) &= \sqrt{n}(\frac{1}{n} X^T X)^{-1} (\frac{1}{n} X^T u) - \text{this is because } (\frac{1}{n})^{-1} = n\end{aligned}$$

We already had that $(X^T X)^{-1} = (\sum_{i=1}^n x_i x_i^T)^{-1}$, and we also have $X^T u = \sum_{i=1}^n x_i u_i$. Therefore, the formula is proved.

(d) We have that $\hat{u} = \mathcal{M}_{\mathbf{X}} y = \mathcal{M}_{\mathbf{X}} (X\beta + u) = \mathcal{M}_{\mathbf{X}} X\beta + \mathcal{M}_{\mathbf{X}} u$.

First, we want to prove that $\mathcal{M}_{\mathbf{X}} X\beta = 0$.

$$\begin{aligned}\mathcal{M}_{\mathbf{X}} X\beta &= [I - X(X^T X)^{-1} X^T] X\beta \\ \mathcal{M}_{\mathbf{X}} X\beta &= X\beta - X(X^T X)^{-1} X^T X\beta \\ \mathcal{M}_{\mathbf{X}} X\beta &= X\beta - XI\beta \\ \mathcal{M}_{\mathbf{X}} X\beta &= 0\end{aligned}$$

Therefore, $\hat{u} = \mathcal{M}_{\mathbf{X}} X\beta + \mathcal{M}_{\mathbf{X}} u = \mathcal{M}_{\mathbf{X}} u$.

Hence, $\sum (\hat{u}_i)^2 = \hat{u}^T \hat{u} = (\mathcal{M}_{\mathbf{X}} u)^T (\mathcal{M}_{\mathbf{X}} u) = u^T (\mathcal{M}_{\mathbf{X}})^T \mathcal{M}_{\mathbf{X}} u = u^T \mathcal{M}_{\mathbf{X}} u$.

The last step is from the fact that $\mathcal{M}_{\mathbf{X}}$ is symmetric, therefore, $(\mathcal{M}_{\mathbf{X}})^T = \mathcal{M}_{\mathbf{X}}$. $\mathcal{M}_{\mathbf{X}}$ is also idempotent, so $\mathcal{M}_{\mathbf{X}} \mathcal{M}_{\mathbf{X}} = \mathcal{M}_{\mathbf{X}}$.

Q8

A function that satisfies the specifications is below:

```
1  ### This function solve a linear system of equations:  $Ax = b$ 
2  # A: n x n square matrix; b: n x 1 vector
3  # If A is singular, print out "It does not have a unique
   solution"
4  # If A is not singular, output the solution
5
6  MC_solve = function (A, b) {
7
8     ### Calculate det_A which is the determinant of A
9     #Note: round to 10 decimal place, because even if A is
       singular,
10    #det(A) may be a very small number not equal to 0 (because
       of R's algorithm)
11    det_A = round(det(A), 10)
12
13    ### Check if det_A is equal to 0, if not, solve the
       equation
14    if (det_A == 0){
15        print("It does not have a unique solution")
16    } else {
17        solution = solve(A, b)
18        solution
19    }
20 }
21
22 ### For problem (a)
23 A1 = matrix(c(1, -2, 1, 0, 2, -8, -4, 5, 9), nrow = 3, byrow
       = TRUE)
24 b1 = c(0, 8, -9)
25
26 # Call the function on A1, b1
27 MC_solve(A1, b1)
28 ### Result: (29, 16, 3)
29
30 ### For problem (b)
31 A2 = matrix(c(1, 4, 2, 2, 5, 1, 3, 6, 0), nrow = 3, byrow =
       TRUE)
32 b2 = c(1, -2, 3)
33
34 # Call the function on A2, b2
35 MC_solve(A2, b2)
36 ### Result: print "It does not have a unique solution"
```